Open-Set Domain Adaptation for Hyperspectral Image Classification Based on Weighted Generative Adversarial Networks and Dynamic Thresholding

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***Abstract*— Recent studies have shown that the deep domain adaptation (DA) technique has achieved remarkable results in cross-domain hyperspectral image (HSI) classification task. However, these DA methods assume that the source and target domains share the same classes, which may not hold true in real-world applications. Under open-set conditions, since the target domain may contain classes unseen in the source domain, direct domain alignment can lead to negative transfer phenomena. Moreover, the presence of multiple unknown classes in the target domain makes it difficult to learn more discriminative classification boundaries between known and unknown classes. To address these issues, we propose an open-set DA (OSDA) method for HSI classification based on weighted generative adversarial networks and dynamic thresholding (WGDT). First, we introduce a class anchor (CA) strategy to learn the metric space of known classes in the source domain. By calculating the similarity between the target-domain samples and the CA, we compute the reliability weights of the samples belonging to known classes. Then, based on these weights, we design an instance-level weighted-domain adversarial learning strategy to better align samples that are more likely to belong to known classes, avoiding negative transfer phenomena. Finally, we propose a dynamic thresholding method to learn the classification boundaries between known and unknown classes in the feature space and reject unknown class samples, thereby separating known class samples in the target domain. The experimental results on four cross-scene HSI classification tasks demonstrate that our proposed method outperforms some existing methods. The code is available at https://github.com/Li-ZK/WGDT.**

***Index Terms*— Class anchor (CA), deep neural network, hyperspectral image (HSI) classification, open-set domain adaptation (OSDA), unsupervised learning.**

I. Introduction

In RECENT years, research on hyperspectral images (HSIs) has continued to advance, and it has been widely used in environmental and disaster monitoring, precision agriculture, and many other fields [1], [2], [3], [4], [5]. HSI contains rich spectral and spatial information, thus providing the possibility for accurate classification of ground objects. Researchers have conducted extensive research on HSI classification, such as 3-D convolutional neural network [6], spectral–spatial residual network [7], and transformer network [8].

Deep learning excels in image classification but assumes data are i.i.d. Real-world variations in acquisition environment conditions or/and sensors cause distribution shifts, hurting generalization. Unsupervised domain adaptation (UDA) helps models adapt to target data without needing target domain labels. Traditional UDA methods can generally be divided into instance-based methods [9], [10], [11], feature-based methods [12], [13], [14], and classifier-based methods [15], [16]. Deep UDA methods can be divided into discrepancy-based [17], [18], [19], adversarial-based [20], [21], [22], and others [23], [24].

However, these DA methods all assume that the source and target domains share the same labels, known as closed-set DA [3]. This closed-set assumption may be invalid, as in real-world application scenarios, the target domain is very likely to contain classes that do not exist in the source domain. Therefore, open-set DA (OSDA) is a more realistic setting [25]. In this article, we mainly follow the setting proposed by Saito et al. [26], with the assumption that the target domain contains all the classes in the source domain and further includes target-specific classes.

In classic DA, domain distribution alignment has always been a key issue. Under open-set conditions, the target domain may contain unknown classes that the source domain has never seen before. The appearance of unknown class data poses more challenges to distribution alignment, and may even

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lead to worse model performance after adaptation, resulting in negative transfer. Therefore, OSDA models need to accurately distinguish known and unknown classes, and only align the data of known classes during DA.

Researchers have made attempts in OSDA. Saito et al. [26] proposed an open-set backpropagation (OSBP) method based on reverse propagation, using a generative model to solve the DA problem. This method does not explore any domain information, but enforces a hard threshold, which may encourage the negative transfer of target samples. Liu et al. [27] proposed a separate to adapt (STA) method that relies solely on the decision of the source training classifier to separate unknown target samples, which may give uncertain predictions for the target samples. Shermin et al. [28] proposed a new adversarial DA model with multiple auxiliary classifiers (DAMCs). This model introduces a weighting module that assigns weights to target samples to represent how likely they are to belong to known and unknown classes, in order to encourage positive transfer during adversarial training while reducing the domain gap between the shared classes of the source and target domains. Chang et al. [29] proposed a deep mutual learning method of mutual separation (MTS), which alternately trains a sample separation network and a distribution matching network to separate unknown samples and match the interdomain distribution. Jang et al. [30] proposed unknown-aware domain adversarial learning (UADAL), which aligns the known distributions of source and target, and separates the unknown target distribution in the feature alignment process. Ru et al. [31] proposed open-set moving-threshold estimation and gradual alignment (OMEGA) to address the problems of intradomain class imbalance and interdomain label shift. Li et al. [32] proposed adjustment and alignment (ANNA) for unbiased OSDA, used front-door adjustment to find the novel-class hidden in the base-class image to correct the biased learning in the source domain, and used decoupled causal alignment to achieve unbiased model transfer. Wan et al. [33] focused on knowledge transfer without access to source-domain data and proposed open-set source-free DA (OS-SFDA). An unknown diffuser is used to explore a more accurate class space. Reliable known knowledge and clustering pseudo-labels are used to achieve more effective known knowledge transfer and unknown generalization. Choe et al. [34] proposed OSDA for semantic segmentation (OSDA-SS), which uses boundary and unknown shape-aware methods to accurately predict the exact boundaries and shapes of unknown classes. Wang et al. [35] proposed the open-set locality-preserving projection (OSLPP) algorithm to learn a discriminative shared subspace between the source and target domains. It progressively selects and rejects pseudo-labeled samples to achieve reliable classification of known classes and recognition of unknown classes.

Current OSDA methods mainly use the image feature space derived from backbone networks and estimate the output probability of the classifier to determine whether the target-domain samples belong to unknown classes. This method assumes that the known classes are tightly clustered in the feature space, while the unknown classes are distant from these clusters. However, unknown class samples may be similar to known class samples, and models trained on known classes may over-confidently misclassify unknown class samples as known classes, which can adversely affect the subsequent domain distribution alignment and make it difficult to suppress negative transfer. In addition, current OSDA methods mainly use the trained $N + 1$-dimensional classifier (N is the number of source-domain classes) to simultaneously classify the known and unknown classes in the target domain. These methods treat all unknown classes in the target domain as one class, but ignore the diversity and complexity. When the unknown class samples have similar features to the known class samples, the model may be difficult to learn more discriminative classification boundaries.

To address the above issues, this article proposes an OSDA method for HSI classification based on weighted generative adversarial networks and dynamic thresholding (WGDT). First, a class anchor (CA) strategy is constructed to learn the metric space of known classes in the source domain. By measuring the similarity between target-domain samples and source-domain CAs, the reliability weight of target-domain samples belonging to known classes can be calculated. Based on this weight, an instance-level weighted-domain adversarial learning strategy is designed for target-domain data. This can better align the samples more likely to known classes and avoid negative transfer during feature alignment. Finally, a dynamic thresholding method is designed to learn the classification boundary between known and unknown classes in the feature space. As a result, unknown class samples can be rejected, and the known class samples in the target domain can be recognized. Our proposed WGDT effectively mitigates domain differences while identifying unknown classes during target-domain classification, which has not been previously explored in the HSI field. The contributions of this article can be summarized as follows.

1) We develop a weight calculation method based on CAs to evaluate the probability of target-domain samples falling into known classes, and thus obtain the weight of target-domain samples belonging to known classes.
2) We design an instance-level weighted-domain adversarial learning strategy that aligns the data distribution between different domains based on different weights of the target sample belonging to a known class, which can better alleviate the problem of cross-domain negative transfer.
3) A learnable dynamic thresholding method is proposed to automatically determine the decision boundaries of known classes in the target domain throughout the training process. This enables the model to more precisely classify samples into either known or unknown classes.

The remainder of this article is organized as follows. Section II introduces relevant concepts of the DA and the open set. Section III introduces the WGDT method in detail. The experimental results on four tasks consisting of eight HSI datasets are shown in Section IV. Finally, a conclusion is drawn in Section V.

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II. Related Work

A. Unsupervised Domain Adaptation

UDA is a DA method that solves the distribution discrepancy between the source and target domains. The goal is to reduce the distribution discrepancy by utilizing labeled data in the source domain and unlabeled data in the target domain, thereby improving the generalization performance of the model in the target domain.

For instance-based methods, the aim is to align the distributions of the source and target domains by adjusting the marginal distribution of the source or target samples. Huang et al. [9] took into account the problem of sample selection bias when training data and testing data come from different distributions, and proposed the nonparametric kernel mean matching (KMM) algorithm, which directly generates resampling weights by matching the distributions of the training and testing sets in feature space without estimating the distribution. Yaras et al. [10] proposed a randomized histogram matching (RHM) data enhancement technique, which improves the generalization ability of deep neural networks on new image sets by applying random spectral transformations to training data. Especially when the new image set is significantly different from the training data in terms of geographic, sensing, or atmospheric conditions, the model is still relatively robust. Li et al. [36] proposed the cost-sensitive self-paced learning (CSSPL) framework, which first uses a simple change detection method to generate pseudo-changed or unchanged labels. Then, it gradually trains from simple samples to complex samples through self-adjusting learning, and automatically assigns different sample weights, thereby selecting more reliable samples for training.

For feature-based methods, the source-domain data and the target-domain data are used to find a common feature space through some transformation, so that the data distribution in the two domains is more similar in this space. Sun et al. [37] focused on how to transfer semantic scene knowledge learned from ground view images to high-resolution remote sensing images and proposed the transfer sparse subspace analysis (TSSA). Jia et al. [38] proposed a gradient feature-oriented 3-D DA (GF-3DDA) approach for HSI classification, using a 3-D Gabor filter to remove noise from the original data and extending the 2-D Sobel gradient (SG) and 2-D derivative-of-Gaussian (DtG) to the 3-D domain to accommodate the spatial–spectral organization of HSIs. Zhou and Shi [39] proposed a deep adapted feature alignment method for UDA. Ren et al. [40] proposed a modified cycle generative adversarial network (CycleGAN) to achieve a higher accuracy and stability.

Classifier-based DA methods mainly focus on adjusting or retraining classifiers to reduce distribution differences between the source and target domains. Zhang et al. [41] proposed the OSDA algorithm via exploring transferability and discriminability (OSDA-ETD) for remote sensing image scene classification, which can solve the problem that novel classes may appear in the target domain. Liu et al. [42] proposed extreme learning machine-based ensemble transfer learning (TL-ELM), which combines extreme learning machine and ensemble learning to improve the accuracy and generalization ability of HSI classification.

For deep DA methods, Zhu et al. [43] proposed deep subdomain adaptation network (DSAN) to expand the feature representation capabilities of deep adaptation networks by aligning the distribution of relevant subdomains in multiple domain-specific layers. Li et al. [44] proposed a two-stage deep DA method for HSI classification to learn a more discriminative deep embedding space and minimize the data difference between the two domains. Huang et al. [45] developed the two-branch attention adversarial DA network for HSI classification. Li et al. [46] considered the separability of the source-domain and target-domain data themselves, and proposed supervised contrastive learning-based UDA (SCLUDA), which performs supervised contrastive learning in the source domain and target domain simultaneously, to reduce the within-class distance and increase the between-class distance.

B. Open Set

Open-set classification methods are mainly divided into discriminative model-based methods and generative model-based methods [25].

For discriminative model-based methods, Scheirer et al. [47] define the open-set classification problem as a constrained minimization problem and add open space risk terms in the modeling process to evaluate the risk of samples belonging to known classes. Zhang and Patel [48] proposed a sparse representation-based open-set recognition. Bendale and Boult [49] proposed the nearest nonoutlier algorithm for open-set classification based on the nearest class mean classifier. Rudd et al. [50] proposed an extreme value machine based on marginal distribution and from a theoretical perspective. Bendale and Boult [51] introduced deep networks into open-set recognition for the first time and proposed OpenMax, calibrating the SoftMax probability for each class using a Weibull distribution model. Yoshihashi et al. [52] proposed classification-reconstruction learning for open-set recognition (CROSR), which uses the difference between the reconstruction result and the original sample to determine whether the testing sample belongs to a known class or an unknown class. Chen et al. [53] proposed the concept of reciprocal points, which is used to represent the extra-class space of each known class. Samples are classified into known or unknown by the difference with reciprocal points. Liu et al. [54] proposed a multitask deep learning method for open-set classification of HSI, which simultaneously learned classification and reconstruction during training. And based on the extreme value theory, the identification of unknown samples is achieved. Yue et al. [55] proposed spectral–spatial latent reconstruction for open-set HSI, which simultaneously learned spectral feature reconstruction, spatial feature reconstruction, and pixel-level classification to obtain a more robust open-set classification model. Sun et al. [56] proposed using a spectral–spatial multiple layer perceptron (MLP)-like network to extract more discriminative features and introduced reciprocal points learning to classify between known and unknown.

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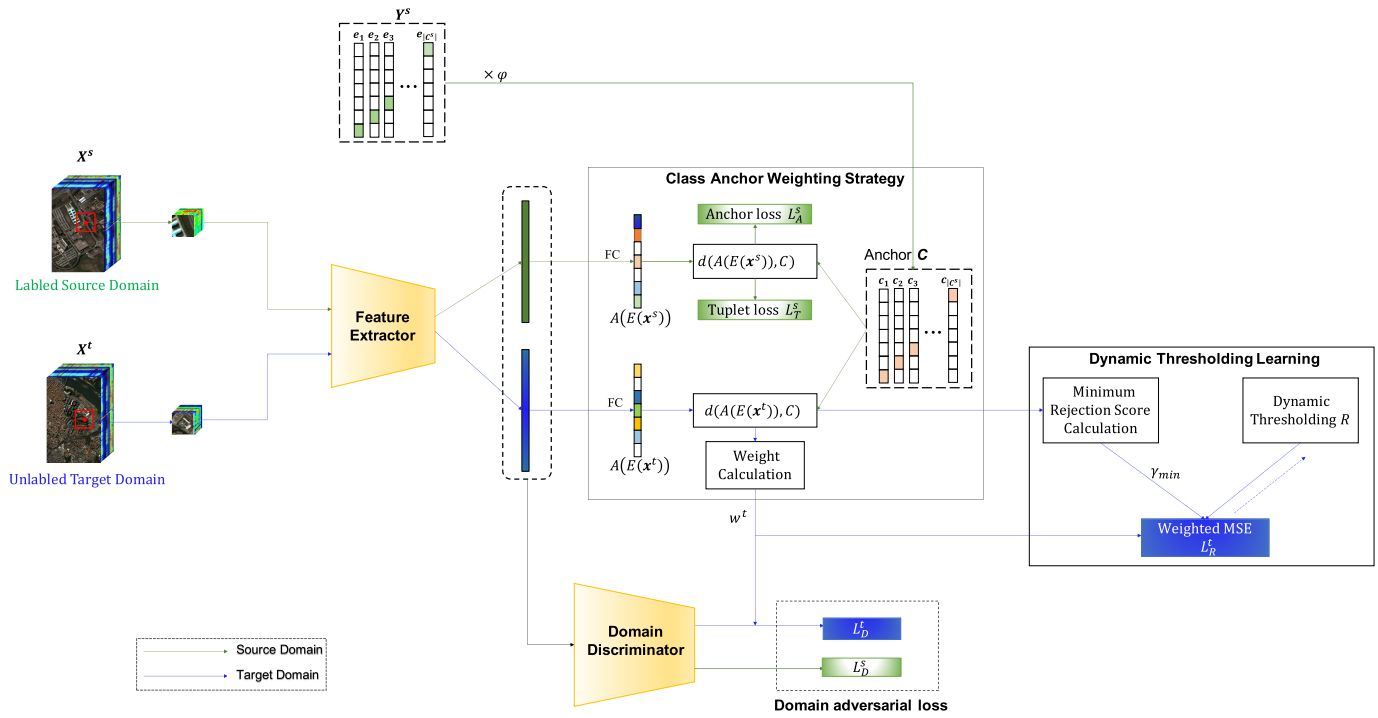


Fig. 1. Framework of the proposed WGDT.

For generative model-based methods, Ge et al. [57] extended OpenMax to Generative OpenMax (G-OpenMax) by introducing generative adversarial networks (GANs), in which the generator generates synthetic samples of new classes and the discriminator learns explicit representations of unknown classes. Neal et al. [58] adopted a similar strategy and proposed counterfactual image generation technology. This technology reformulates open-set recognition as a classification with an additional class, and constrains the generation process so that the sample does not belong to a known class, but at the same time, it can continuously approach the known class. Jo et al. [59] also used GAN to generate fake data as unknown class data, so that the model has a certain understanding of the unknown class. Yu et al. [60] proposed an adversarial sample generation framework, which can not only generate unknown class data as negative samples, but also generate positive samples based on seen classes.

Recently, few-shot open-set was proposed to solve the problem of few training samples of known classes in the open-set setting. Pal et al. [61] proposed a few-shot open-set recognition of HSIs with an outlier calibration network, using outlier calibration network to reject unknown and introducing a feature enhancement paradigm to estimate the distribution of known classes. Furthermore, Pal et al. [62] proposed a meta-learning-based GAN open set recognition method to simultaneously generate pseudo-known classes and unknown classes to achieve a more refined separation of known and unknown classes. Wang et al. [63] proposed an energy-based hybrid model, which uses the class similarity and pixel similarity of query samples and the support set to learn the energy score, to achieve robust few-shot open-set recognition.

III. Proposed Method

In OSDA, given a source domain $D^s = \{(X^s, Y^s)\} = \{(x_i^s, y_i^s)\}_{i=1}^{n^s}$ and a target domain $D^t = \{X^t\} = \{x_j^t\}_{j=1}^{n^t}$, where n^s is the number of labeled samples and n^t is the number of unlabeled samples. A set of classes in the source domain and target domain are denoted as C^s and C^t, respectively, and $C^s \subset C^t$. The class set $C^{t\backslash s}$ in the target domain that does not belong to the source domain represents unknown classes. The data distributions of the source domain and target domain are p and q, respectively, and $p \neq q$.

The structure of the proposed OSDA classification model for HSI is shown in Fig. 1. First, the data x^s and x^t from the source and target domains are input into the feature extractor E to obtain the corresponding features $E(x^s)$ and $E(x^t)$. Then, both features are input into the CA weighting module and domain discriminator D. For the source-domain features $E(x^s)$, the classification loss in the CA weighting module and the domain discrimination loss in the domain discriminator are calculated, respectively. For the target-domain features $E(x^t)$, by calculating the distance from the CA, we obtain the reliability weight w^t that the sample belongs to the known classes, so as to weight the domain discrimination loss of the target-domain sample at the instance level to avoid negative transfer.

A. Spectral–Spatial Feature Extractor

The feature extraction network is based on the dual-channel residual network, including the spectral feature extractor, the spatial feature extractor, and the feature fusion, as shown in Fig. 2. We use a cube of size $7 \times 7 \times 48$ (height, width, and the number of bands, respectively) as an input example to illustrate the structure of the feature extractor. The feature

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extractor consists of multiple convolution operations, and each convolution operation is followed by a batch normalization (BN) layer and an ReLU layer.

1) *Spectral Feature Extraction:* First, the sample is processed by 24 convolution kernels of size $1 \times 1 \times 7$, and then, convolution is performed on the spectral dimension through a residual block composed of two convolution layers. Finally, 192 convolution kernels of size $1 \times 1 \times 21$ are used to process the intermediate features into 192 2-D spatial blocks of size 7×7, which are spectral features.
2) *Spatial Feature Extraction:* First, the sample is processed by 24 convolution kernels of size $1 \times 1 \times 48$ to obtain 24 2-D spatial blocks of size 7×7, and then, convolution is performed on the spatial dimension through a residual block composed of two convolution layers to obtain 96 spatial features of size 7×7.
3) *Feature Fusion:* After obtaining the spectral and spatial features, they are concatenated along the channel dimension to obtain an intermediate feature of size $7 \times 7 \times 288$. Then, an average pooling layer of size 7×7 is used to process the intermediate features to obtain a feature vector of length 288.

B. CA Weighting Strategy

The CA strategy is extended for accurate classification of target-domain samples and estimates the similarity between target-domain samples and known classes of the source domain, providing instance-level reliability weights for domain adversarial learning losses. This avoids the negative transfer phenomenon caused by label mismatch between the source domain and the target domain during the alignment process.

We assign each known class in the source domain its own anchor, and the one-hot is scaled by a factor of φ. This will expand the size of the feature space and reserve more space for unknown sample features. The formula can be expressed as

$$\boldsymbol{C} = \left(\boldsymbol{c}_1, \ldots, \boldsymbol{c}_{|C^s|}\right) = \left(\varphi \cdot \boldsymbol{e}_1, \ldots, \varphi \cdot \boldsymbol{e}_{|C^s|}\right) \tag{1}$$

where $\boldsymbol{C}$ is the class anchor, the one-hot and anchor of the ith class are $\boldsymbol{e}_i$ and $\boldsymbol{c}_i$, respectively, and $|C^s|$ is the number of known classes in the source domain.

Then, the feature $E(x)$ extracted by the feature extractor E passes through a fully connected layer and is mapped to a vector with a length $|C^s|$, expressed as $A(E(x))$. Calculate the distance between the vector and each CA

$$\begin{aligned} &d(A(E(x)), \boldsymbol{C}) \\ &= (||A(E(x)) - \boldsymbol{c}_1||_2, \ldots, ||A(E(x)) - \boldsymbol{c}_{|C^s|}||_2)^{\mathrm{T}}. \end{aligned} \tag{2}$$

A sample x^s in the source domain, labeled y^s, use anchor loss to minimize the distance of its features to true CAs, and use tuplet loss to maximize the distance to other CAs

$$L_A^s(x^s, y^s) = \left|\left|A\left(E\left(x^s\right)\right) - \boldsymbol{c}_{y^s}\right|\right|_2 \tag{3}$$

$$L_T^s(x^s, y^s) = \log\left(1 + \sum_{j \neq y^s}^{|C^s|} e^{d_{y^s} - d_j}\right) \tag{4}$$

where d_{y^s} is the distance between $A(E(x^s))$ and the true CA, and d_j is the distance from the anchor of class j.

For the target-domain sample x^t, the distance of $A(E(x^t))$ to the anchor is used to evaluate the similarity to the known class, especially, the minimum distance l between $A(E(x^t))$ and each CA point. The smaller l indicates that the sample is more similar to the known class. Then, in order to achieve reasonable weighting of domain adversarial loss and dynamic thresholding loss, l is normalized to between (0, 1). Since l has the opposite meaning to weight representation, the normalization method is modified here. The formula is expressed as follows:

$$l = \min d\left(A\left(E\left(x^t\right)\right), \boldsymbol{C}\right) \tag{5}$$

$$w^t = \frac{l_{\max} - l}{l_{\max} - l_{\min} + \epsilon} \tag{6}$$

where $l_{\max}$ and $l_{\min}$ are the maximum and minimum values of L in a batch, respectively, w^t is the reliability weight of the target-domain sample x^t belonging to the known classes, and ϵ is used to prevent the denominator from being zero. A larger w^t indicates that the x^t is more similar to the known classes.

C. Instance-Level Weighted-Domain Adversarial Learning

Ganin et al. [64] proposed to use domain adversarial neural networks to align the feature distributions of the two domains. The domain adversarial loss formula is as follows:

$$\begin{aligned} L_{\mathrm{DANN}} = &\frac{1}{N^s}\sum_{i=1}^{N^s} L_{\mathrm{bce}}\left(D\left(E\left(x_i^s\right)\right), d_i^s\right) \\ &+ \frac{1}{N^t}\sum_{i=1}^{N^t} L_{\mathrm{bce}}\left(D\left(E\left(x_i^t\right)\right), d_i^t\right) \end{aligned} \tag{7}$$

where d is the domain label, N^s is the number of source-domain samples, and N^t is the number of target-domain samples.

If the unknown class samples in the target domain are aligned with the known classes in the source domain, the negative transfer will occur, that is, the classification performance of the model on the target domain will decrease, and it is prone to cause confusion between known and unknown classes. Therefore, we can use w^t to perform instance-level weighting on the target-domain adversarial loss, so that samples with higher reliability play a greater role in the alignment process, while samples with lower reliability participate less, thereby mitigating the negative transfer phenomenon caused by alignment. The framework of the domain discriminator is shown in Fig. 3.

Source-Domain Adversarial Loss: For the source-domain samples, it must come from known classes. Therefore, the domain adversarial loss of source-domain samples is reliable and all samples will participate in the alignment, which can be expressed as

$$L_D^s = \frac{1}{N^s}\sum_{i=1}^{N^s} L_{\mathrm{bce}}\left(D\left(E\left(x_i^s\right)\right), d_i^s\right). \tag{8}$$

Target-Domain Adversarial Loss: Since the samples in the target domain have no labels, it cannot be determined whether they belong to known classes. If samples of unknown classes are included in the alignment, it will cause a negative transfer,

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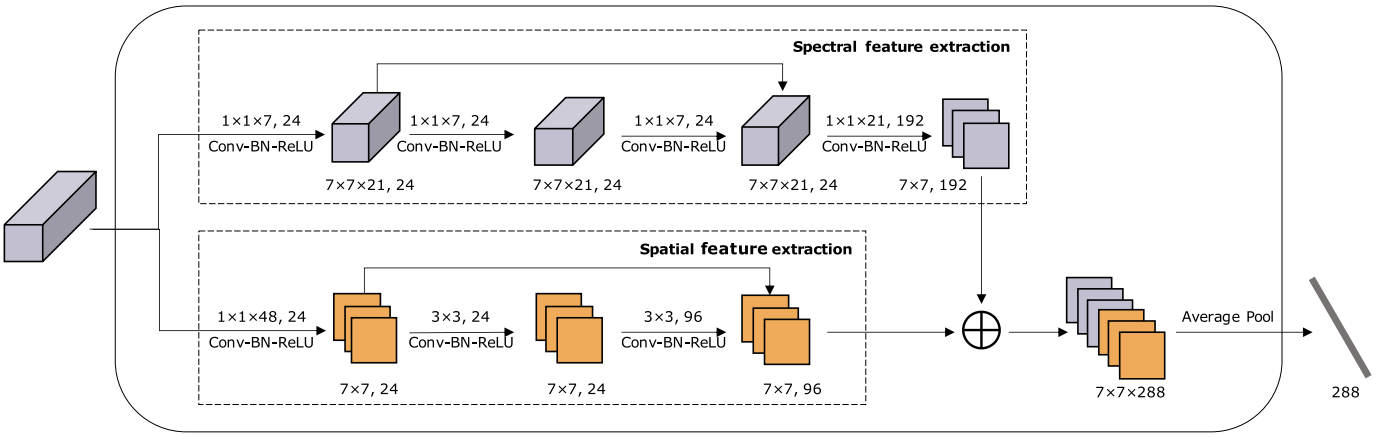


Fig. 2. Feature extractor has two channels, extracting spectral and spatial features, respectively, and finally merging them into a vector.

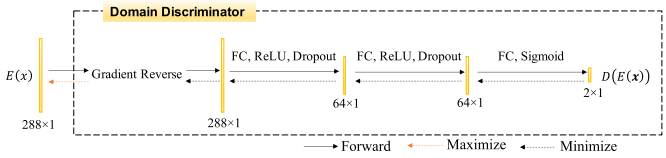


Fig. 3. Framework of the domain discriminator.

reduce the classification performance of the model in the target domain, and confuse unknown classes with known classes, resulting in a reduction in the accuracy of unknown class recognition. Therefore, the target-domain adversarial loss is weighted using the reliability weight of the sample belonging to a known class provided by the CA weighting strategy, which can be expressed as

$$L_D^t = \frac{1}{N^t}\sum_{i=1}^{N^t} w_i^t L_{\text{bce}}\left(D\left(E\left(x_i^t\right)\right), d_i^t\right) \tag{9}$$

where w^t is the reliability weight provided by the CA strategy.

Therefore, the total domain adversarial loss is

$$L_D = L_D^s + L_D^t. \tag{10}$$

D. Dynamic Thresholding Learning

Manually setting thresholds to classify known classes and unknown classes may only be effective for certain data, and it is difficult to determine an optimal threshold for accurate classification. Therefore, a method of dynamically determining the threshold is designed, which can learn to estimate the decision boundary of known classes in the target domain during the training process of the model, providing a reliable basis for rejecting unknown classes. Specifically, the proposed method sets a learnable parameter R for learning the known class feature boundaries in the target domain. If the rejection score of the sample is greater than R, it is assigned to the unknown class.

First, calculate the rejection score vector γ and obtain the minimum rejection score $\gamma_{\min}$, which can be expressed as

$$\gamma = d \circ (1 - \text{softmin}(d)) \tag{11}$$

$$\gamma_{\min} = \min(\gamma) \tag{12}$$

where $\circ$ is the elementwise product, and d is a vector with length $|C^s|$, representing the distance between the target-domain sample features and each CA.

A simple idea is to use the mean-squared error loss function to reduce the distance between R and $\gamma_{\min}$. By minimizing this loss function, R can be brought close to the boundary of the rejection score

$$L_{\text{MSE}} = \frac{1}{N^t}\sum_{i=1}^{N^t}\left(R - \text{sg}\left[\gamma_{\min}\right]\right)^2 \tag{13}$$

where N^t is the number of target-domain samples and sg[·] is the stop-gradient operator.

However, this method is not suitable for OSDA because directly using minimizing the mean-squared error loss to update the boundary R, unknown class samples will be included. As a result, the value of the boundary R is too large, which may cover all target-domain sample features and cannot effectively reject unknown classes. Therefore, it is necessary to learn only the boundaries of known class sample features in the target domain as much as possible, while ignoring unknown class sample features.

To this end, the mean-squared error loss is modified here. The corresponding loss is weighted using the reliability weight w that the target-domain sample belongs to the known class given by the CA weighting module. The formula is as follows:

$$L_R^t = \frac{1}{N^t}\sum_{i=1}^{N^t} w_i^t\left(R - \text{sg}\left[\gamma_{\min}\right] - m\right)^2 \tag{14}$$

where m is a hyperparameter that can be used to control the distance between R and $\gamma_{\min}$.

E. Training and Testing Processes

The proposed method consists of two stages in the training process. The first stage is the pretraining stage, which is used to separate known and unknown classes. The second stage is to align the source and target domains and learn the threshold.

1) *The First Stage:* The feature extractor E and the CA weighting module A are trained using source-domain samples to classify known classes and leave room for

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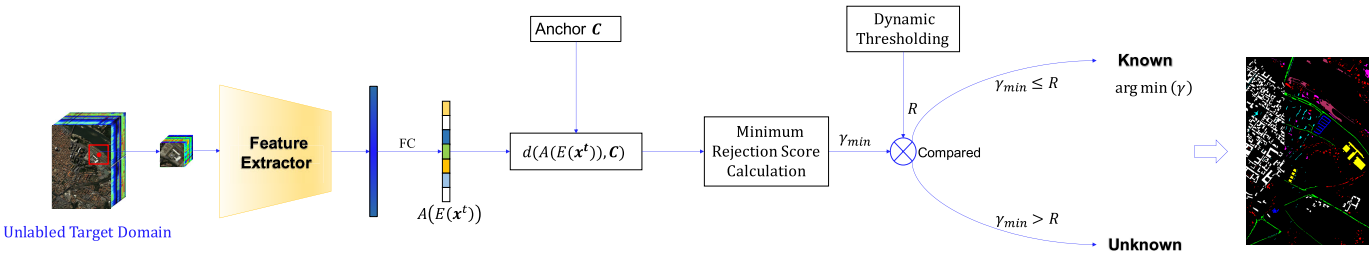


Fig. 4. Classification flowchart of testing stage.

Algorithm 1 Training of WGDT

Input:

Source domain dataset $D^s = \{(x_1^s, y_1^s), \ldots, (x_{n^s}^s, y_{n^s}^s)\}$.
Target domain dataset $D^t = \{x_1^t, \ldots, x_{n^t}^t\}$.
Hyperparameters α, β, λ.
The first stage and second stage of training epochs epoch[1] and epoch[2].
Randomly initialized model parameters θ_E, θ_A, θ_D and θ_R.

Output:

Model parameters after training.

```
Initialize the CA C
for epoch in epoch[1] :
    for batch in dataloader:
        Extracting source domain features E(x^s)
        Send E(x^s) into the CA module and get A(E(x^s))
        Calculate the distance vector d(A(E(x^s)), C)
        Calculate source domain loss L_A^s and L_T^s
        Update model parameters θ_E, θ_A
for epoch in epoch[2] :
    for batch in dataloader:
        Extracting features E(x^s) and E(x^t)
        Class module calculates A(E(x^s)) and A(E(x^t))
        Calculate the distance vector d(A(E(x^s)), C)
        and d(A(E(x^t)), C)
        Calculate source domain loss L_A^s and L_T^s
        Calculate reliability weights w^t
        Calculate the min(γ) of the target domain features
        Calculate dynamic thresholding loss L_R^t
        Domain adversarial module calculates D(E(x^s))
        and D(E(x^t))
        Calculate domain adversarial loss L_D
        Update model parameters θ_E, θ_A, θ_D and θ_R
```

Algorithm 2 Testing of WGDT

Input:

Target domain dataset $D^t = \{x_1^t, \ldots, x_{n^t}^t\}$.
Trained model parameters θ_E, θ_A, and θ_R.

Output:

Predicted values of the target domain dataset.

```
for batch in dataloader:
    Extracting target domain features E(x^t)
    Send E(x^t) into the CA module and get A(E(x^t))
    Calculate the distance vector d(A(E(x^t)), C)
    Calculate the min(γ) of the target domain features
    If min(γ) > R:
        ŷ = unknown
    else:
        ŷ = arg min(γ)
```

unknown features. Use θ_E and θ_A to represent the parameters of E and A, respectively. The cost function to optimizing parameters $\hat{\theta}_E$ and $\hat{\theta}_A$ can be expressed as

$$\left(\hat{\theta}_E, \hat{\theta}_A\right) = \arg\min_{\theta_E, \theta_A} L_T^s + \alpha L_A^s. \tag{15}$$

2) *The Second Stage:* The feature extractor E and the domain discriminator D are trained using source and target domain samples to align the features of the two domains, and learn the boundary R in the target domain to make decisions on whether the sample belongs to a known class or an unknown class. Meanwhile, continue to use source-domain samples to train the CA weighting module A to retain the knowledge from the source domain. The total loss can be expressed as

$$L_{\text{total}} = L_T^s + \alpha L_A^s - \beta L_D + \lambda L_R^t. \tag{16}$$

Use θ_D and θ_R to represent the parameters of the domain discriminator D and the boundary R, respectively, and the optimized parameters are represented by $\hat{\theta}_D$ and $\hat{\theta}_R$. The overall optimization objective is as follows:

$$\left(\hat{\theta}_E, \hat{\theta}_A, \hat{\theta}_R\right) = \arg\min_{\theta_E, \theta_A, \theta_R} L_{\text{total}} \tag{17}$$

$$\left(\hat{\theta}_D\right) = \arg\max_{\theta_D} L_{\text{total}}. \tag{18}$$

The first stage separates known and unknown classes so that the model has some understanding of the known classes. This can provide a more reliable similarity weight and avoid the phenomenon of negative transfer caused by unknown classes participating in excessive weighting due to insufficient knowledge of known and unknown classes in the early stage of model training. Finally, accurate identification of known classes is achieved, and unknown classes are rejected in the target domain.

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TABLE I
NUMBER OF SAMPLES IN THE PU–PC TASK

No.	Class	PU	PC
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare Soil	5029	6584
8	Unknown		42826
	Total	39332	82181

TABLE II
NUMBER OF SAMPLES IN THE HU13–HU18 TASK

No.	Class	HU13	HU18
1	Grass healthy	345	9799
2	Grass stressed	365	32502
3	Trees	365	18616
4	Water	285	266
5	Residential buildings	319	39772
6	Non-residential buildings	408	223752
7	Road	443	45866
8	Unknown		134283
	Total	2530	504856

TABLE III
NUMBER OF SAMPLES IN THE DIONI–LOUKIA TASK

No.	Class	Dioni	Loukia
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Sparse Sclerophyllous Vegetation	6374	2361
8	Rocks and Sand	492	453
9	Coastal Water	398	421
10	Unknown		4788
	Total	11623	10317

TABLE IV
NUMBER OF SAMPLES IN THE ZY–GF TASK

No.	Class	ZY	GF
1	Architecture	451	360
2	Sea	2076	2395
3	Offshore water	1558	1305
4	Unknown		1415
	Total	4085	5475

In the testing stage as shown in Fig. 4, a two-step decision method is adopted to determine the class to which the target-domain sample belongs. If $\min(\gamma)$ is greater than R, the sample is predicted as an unknown class, or the sample is predicted to be the class corresponding to $\min(\gamma)$, i.e.,

$$\hat{y} = \begin{cases} \text{unknown}, & \min(\gamma) > R \\ \arg\min(\gamma), & \min(\gamma) \leq R. \end{cases} \tag{19}$$

IV. EXPERIMENTS

A. Datasets

In order to verify the effectiveness of the proposed method, eight datasets were used for experiments, including Pavia University (PU), Pavia Center (PC), Houston2013 (HU13), Houston2018 (HU18), Dioni, Loukia, Ziyuan1-02D Yancheng (ZY), and GaoFen-5 Yancheng (GF). In the experiments, these eight datasets were combined into four OSDA classification tasks. The first task uses PU as the source domain and PC as the target domain. The second task uses HU13 as the source domain and HU18 as the target domain. The third task uses Dioni and Loukia as the source domain and target domain,

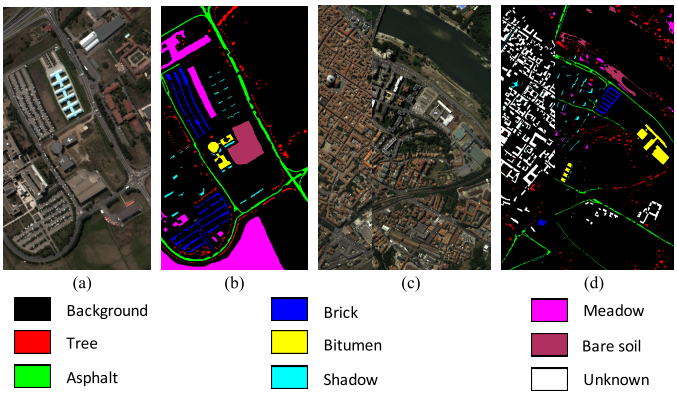


Fig. 5. PU–PC task. (a) Pseudo-color image of PU. (b) Ground-truth image of PU. (c) Pseudo-color image of PC. (d) Ground-truth image of PC.

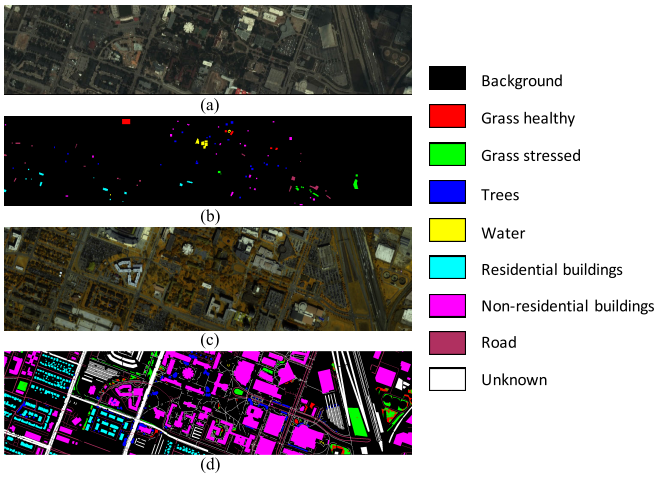


Fig. 6. HU13–HU18 task. (a) Pseudo-color image of HU-13. (b) Ground-truth image of HU-13. (c) Pseudo-color image of HU18. (d) Ground-truth image of HU18.

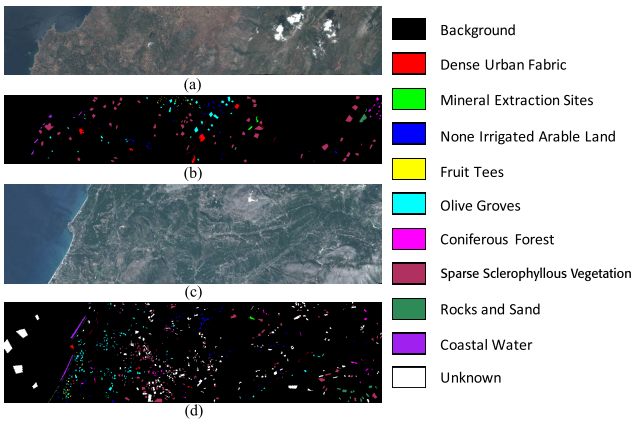


Fig. 7. Dioni–Loukia task. (a) Pseudo-color image of Dioni. (b) Ground-truth image of Dioni. (c) Pseudo-color image of Loukia. (d) Ground-truth image of Loukia.

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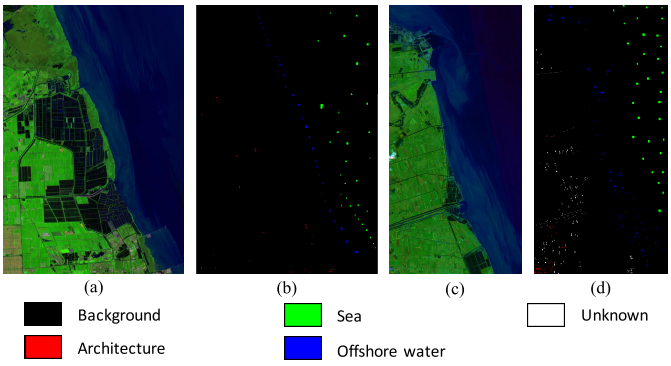


Fig. 8. ZY–GF task. (a) Pseudo-color image of ZY. (b) Ground-truth image of ZY. (c) Pseudo-color image of GF. (d) Ground-truth image of GF.

respectively. The fourth task uses ZY as the source domain and GF as the target domain.

1) *PU–PC:* Pavia is an HSI collected by the ROSIS sensor in Pavia, Northern Italy, which contains two scenes of PU and PC. The spatial size of PU is 610 × 340 pixels, the spatial resolution is 1.3 m, and there are 115 bands in total. After removing the 12 bands that are seriously affected by noise, 103 bands remain. The spatial size of the PC is 1096 × 715 pixels, the spatial resolution is also 1.3 m, and 102 bands remain after processing. In the experiments, 102 bands common to both datasets were used. Select seven common classes as the known classes, and select a class (Tiles) in the target domain as the unknown class. The pseudo-color images and ground truth images of the PU-PC task are shown in Fig. 5. The number of samples in the PU–PC task is shown in Table I.
2) *HU13–HU18:* The Houston dataset contains two scenes, HU13 and HU18. HU13 is an HSI collected by the ITERS CASI-1500 sensor. The collection time was 2013 in Houston, TX, USA, and its surrounding rural areas. The spatial size of the HSI is 349 × 1905 pixels, and the spatial resolution is 2.5 m and contains 144 bands. HU18 is also an HSI collected near Houston. It was a dataset used in the competition held by IEEE GRSS Data Fusion in 2018. The spatial size is 601 × 2384 pixels, the spatial resolution is 1 m, the wavelength range is from 380 to 1050 nm, and there are 48 bands in total. In the experiment, 48 bands common to the two datasets were used. Merge the evergreen trees and deciduous trees classes in HU18 into the trees class, so that the two domains have seven known classes, and select the remaining 12 classes in the target domain as an unknown class. The pseudo-color images and ground truth images of the HU13-HU18 task are shown in Fig. 6. The number of samples in the HU13-HU18 task is shown in Table II.
3) *Dioni–Loukia:* Dioni and Loukia obtained from the Hyperion sensor, carried by the National Aeronautics and Space Administration (NASA) Earth Observing 1 (EO-1) satellite. After processing, 176 bands remain. The spatial size of Dioni is 250 × 1376 pixels, and the spatial size of Loukia is 249 × 945 pixels. In the experiment, nine ground object classes were selected as known classes, and three classes (dense sclerophyllous vegetation, sparsely vegetated areas, and water) in the target domain were selected as an unknown class. The pseudo-color images and ground truth images of the Dioni-Loukia task are shown in Fig. 7. The number of samples in the Dioni-Loukia task is shown in Table III.
4) *ZY–GF:* ZY and GF are acquired by Advanced Hyperspectral Imager (AHSI) on the Chinese Ziyuan1-02D and GaoFen-5 satellites, respectively. The spatial size of ZY is 1398 × 942 pixels, and the spatial size of GF is 1175 × 585 pixels. Both HSIs contain 147 bands. In the experiment, three common classes are selected as known classes, and the remaining four classes in the target domain are selected as unknown classes. The pseudo-color images and ground truth images of the ZY-GF task are shown in Fig. 8. The number of samples in the ZY-GF task is shown in Table IV.

B. Metrics

Following previous work [65], HOS was used to evaluate the performance of the model. HOS is the harmonic mean of the average accuracy of known classes (OS*) and the accuracy of unknown classes (Unk), which takes into account the deficiencies of OS (the average accuracy of all classes) in open-set classification. For example, a classification method is not designed for dealing with unknown classes, and the accuracy of unknown classes is 0.0%, but it has a perfect classification on known classes (assume ten classes), and the OS* can reach 100.0%. Although the method does not identify unknown classes, it shows a high-performance evaluation with an OS of 90.9%. As the number of known classes increases, the overall performance evaluation of the method will be more seriously affected, ignoring the unknown classes. Therefore, HOS is used to evaluate the overall performance of the open-set classification model for known and unknown class classification, which can be expressed as

$$\mathrm{HOS} = 2\frac{\mathrm{OS}^* \times \mathrm{Unk}}{\mathrm{OS}^* + \mathrm{Unk}} \tag{20}$$

where HOS is a high score only if the method performs well on both known and unknown classes. Moreover, it makes the performance independent of the number of known classes.

C. Implementation Details

In the proposed method, all experiments are conducted independently ten times, and the average value and standard deviation are calculated after statistics to reduce the impact of random factors such as weight initialization and data disturbance. In each experiment, 180 labeled samples from each class in the source domain are randomly selected as the training set to train the network. The patch size of data samples is uniformly set to 7 × 7. The training epochs are set to 20 in the first stage and 150 in the second stage, and the batch size is 32. For the optimization of the network, stochastic gradient descent is used, with the weight decay parameter set to $5e^{-4}$ and the momentum parameter set to 0.9. The learning rates of feature extractor E and CA weighting module A are uniformly set to 0.001, and the learning rate of boundary R is uniformly set to 0.0001. The learning rate of the domain discriminator D

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TABLE V
CLASSIFICATION RESULTS ON THE PU–PC TASK

Class	Closed-set			Open-set								
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT
1	96.4	98.0	94.6	93.2	95.6	93.3	80.0	91.0	64.1	71.3	84.6	**96.1**
2	99,8	92.1	96.7	84.8	85.7	77.4	67.9	**87.6**	15.3	72.0	65.5	81.0
3	75.8	80.3	56.7	70.7	**71.7**	68.8	55.0	65.7	39.3	53.7	68.6	47.1
4	17.8	13.8	25.1	63.1	**77.9**	63.7	54.5	75.7	45.7	25.3	65.0	47.0
5	100.0	100.0	100.0	98.9	99.2	98.9	96.1	67.3	61.0	91.7	99.0	**100.0**
6	86.2	71.4	88.3	48.0	39.4	29.4	21.2	**66.8**	55.1	65.8	64.9	64.9
7	83.4	44.1	73.2	44.6	34.3	35.2	38.4	25.8	0.8	18.3	9.2	**46.4**
Unk (%)	0.0	0.0	0.0	34.3	38.6	34.4	25.8	19.8	**98.6**	64.9	65.9	86.3
OS* (%)	79.9	71.4	76.4	71.9	**72.0**	66.7	63.3	68.6	40.2	56.9	65.3	68.9
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	46.3±3.7	49.9±6.6	45.3±3.1	35.4±10.6	30.2±7.4	56.1±12.6	59.2±14.4	65.0±4.7	**76.5±4.6**

TABLE VI
CLASSIFICATION RESULTS ON THE HU13–HU18 TASK

Class	Closed-set			Open-set								
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT
1	48.4	94.6	98.4	84.1	74.1	85.9	81.1	88.3	58.6	47.6	**94.2**	74.5
2	56.5	75.4	42.9	34.6	28.0	31.1	25.7	**43.1**	33.8	37.0	41.6	34.3
3	64.6	60.1	45.4	62.5	56.2	67.8	**74.7**	73.7	36.8	49.2	51.8	48.1
4	83.1	99.9	77.3	39.4	61.1	52.1	79.5	80.8	25.6	43.1	56.3	**90.6**
5	83.9	82.9	77.1	60.5	65.4	65.9	**76.7**	76.5	19.6	35.8	42.3	65.7
6	47.5	82.7	76.3	33.9	32.2	27.9	27.3	42.9	13.0	35.5	27.2	**64.7**
7	70.2	35.3	72.8	52.1	51.9	47.4	40.1	**59.4**	8.5	11.3	36.6	3.7
Unk (%)	0.0	0.0	0.0	26.9	29.6	29.6	24.3	13.2	**89.2**	57.8	56.0	70.2
OS* (%)	64.9	75.8	70.0	52.4	52.7	54.0	57.7	**66.4**	28.0	37.1	50.0	54.5
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	35.0±5.4	37.6±3.7	37.7±4.9	33.1±8.2	21.4±8.1	41.5±12.5	43.7±9.2	51.7±5.6	**61.1±2.8**

varies on different datasets to adapt to different domain gaps. On the PU–PC, the learning rate of the domain discriminator is set to 0.01; on the HU13–HU18, the Dioni–Loukia, and the ZY–GF, the learning rate of the domain discriminator is set to 0.0001.

All experiments in this article were completed on a DELL workstation. For hardware environment information, the central processing unit (CPU) model is Intel Xeon[1] Gold 6226R, the main frequency is 2.90 GHz, and the number of threads is 64. The graphics processing unit (GPU) model is NVIDIA GeForce RTX 3090/PCIe/SSE2, and the graphics memory is 24 GB. The memory size is 224 GB. For software environment information, the system used is Ubuntu 20.04.1 LTS, 64 bit. Compute unified device architecture (CUDA) is 11.3. Using version 3.7.15 of Python. PyTorch version is 1.12.1.

D. Experiments

To evaluate the performance of the proposed method, it is compared with three state-of-the-art DA methods and eight state-of-the-art OSDA classification methods, including CLDA [66], SCLUDA [46], TAADA [45], DACD [67], OSBP [26], DAMC [28], MTS [29], STA [27], UADAL [30], OMEGA [31], and ANNA [32].

1) *PU–PC:* The experimental results of different comparison methods for the PU–PC task are shown in Table V. The results show that the performance of the proposed method is better than other methods and has been greatly improved. Although the first four comparison methods have good classification performance on the known class, it is difficult to correctly classify the unknown class. UADAL achieves a high performance of 98.6% in the recognition of the unknown class, but the classification accuracy of known classes is low. The proposed method has a good performance in both known and unknown classes, with OS*, Unk, and HOS reaching 68.9%, 86.3%, and 76.5%, respectively. HOS improved by 11.5% compared to the other best methods. The classification maps of the proposed and compared methods are shown in Fig. 9.

2) *HU13–HU18:* The experimental results of different comparison methods for the HU13–HU18 task are shown in Table VI. The results show that the performance of the proposed method is superior to other methods. MTS has the best performance in the classification of known classes, with OS* reaching 66.4%, but the classification accuracy of the unknown class is low, Unk is 13.2%, so HOS also has a low evaluation. UADAL has the best performance in the classification of the unknown class, Unk reaches 89.2%, but the classification accuracy of known classes is low, OS* is 28.0%, and HOS is 41.5%. The OS* and Unk of the proposed method reached 54.5% and 70.2%, respectively, and the HOS reached 61.1%. HOS improves by 9.4% compared to other best methods. The classification maps of the proposed and compared methods are shown in Fig. 10.

3) *Dioni–Loukia:* The experimental results of different comparison methods for the Dioni–Loukia task are shown in Table VII. In terms of known and unknown identification, the proposed method has the highest OS* and Unk, which are 46.6% and 52.0%, respectively, and HOS reaches 48.9%. HOS improves by 7.3% compared to other best methods. The classification maps of the proposed method and compared methods are shown in Fig. 11.

4) The experimental results of different comparison methods for the ZY–GF task are shown in Table VIII. DAMC has the best performance in the classification of known classes, with OS* reaching 94.0%, but the Unk is only 14.5%. The OS*

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TABLE VII
CLASSIFICATION RESULTS ON THE DIONI–LOUKIA TASK

Class	Closed-set			Open-set								
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT
1	0.7	4.4	48.4	12.9	**24.8**	19.3	9.5	16.9	4.9	10.8	8.5	3.9
2	100.0	71.1	95.4	48.1	43.5	57.2	68.7	83.9	96.7	**97.2**	87.6	88.0
3	59.4	26.2	72.5	21.7	33.4	31.3	41.5	28.5	28.0	19.3	21.7	**51.5**
4	93.0	100.0	79.9	70.4	72.0	70.4	76.5	**84.1**	55.7	39.4	49.4	56.3
5	0.0	0.1	36.6	8.0	13.1	13.2	**22.3**	7.9	2.5	17.6	2.9	0.1
6	81.3	64.0	56.2	62.8	58.9	56.9	39.8	**73.3**	36.4	38.1	57.7	63.7
7	18.3	16.3	50.0	11.6	**17.1**	11.7	3.5	11.1	3.0	2.8	11.8	5.7
8	6.3	72.1	45.6	3.1	2.0	4.7	7.4	24.7	38.1	10.9	10.3	**52.7**
9	100.0	97.4	99.8	28.5	8.7	18.0	61.8	82.3	85.7	83.6	14.8	**97.7**
Unk (%)	0.0	0.0	0.0	23.4	25.0	23.0	13.9	6.2	48.8	**54.0**	46.4	52.0
OS* (%)	51.0	50.2	64.9	29.7	30.4	31.4	36.8	45.6	39.0	35.5	29.4	**46.6**
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	25.3±5.2	26.4±3.1	25.7±5.3	18.5±10.2	10.3±7.0	41.6±4.3	40.4±10.3	34.0±6.1	**48.9±1.3**

TABLE VIII
CLASSIFICATION RESULTS ON THE ZY–GF TASK

Class	Closed-set			Open-set								
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT
1	100.0	100.0	100.0	85.8	89.2	82.6	72.9	**90.6**	62.3	67.8	29.1	83.7
2	59.2	100.0	100.0	**100.0**	**100.0**	**100.0**	91.6	80.8	15.2	65.4	53.7	99.6
3	97.6	100.0	82.4	93.3	91.6	**99.5**	92.5	75.0	80.7	77.8	14.8	81.2
Unk (%)	0.0	0.0	0.0	13.4	15.5	14.5	49.2	51.0	58.8	48.3	76.4	**90.2**
OS* (%)	85.6	100.0	94.1	93.0	93.6	**94.0**	85.7	82.1	52.7	70.4	32.5	88.2
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	23.4±4.3	26.6±0.7	25.0±1.1	60.4±12.1	62.7±11.9	49.3±9.8	55.7±9.3	43.6±8.2	**88.9±3.8**

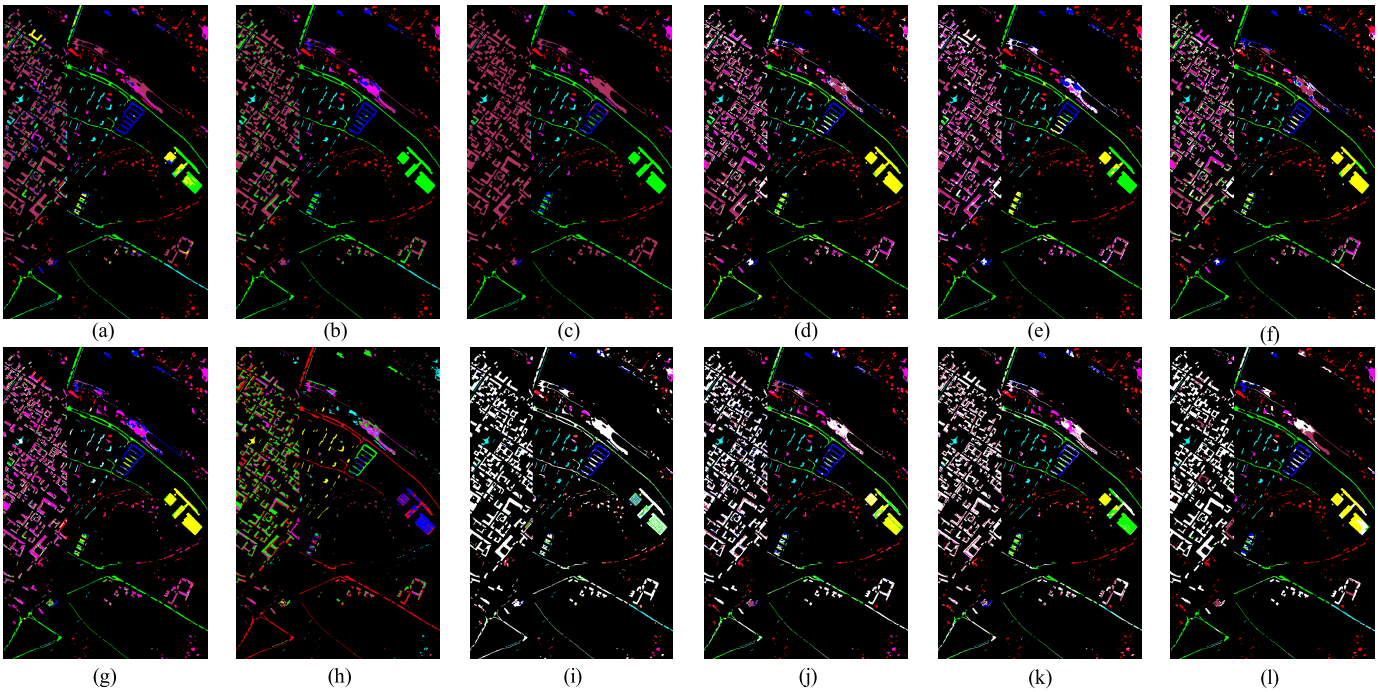


Fig. 9. Classification map of the PU–PC task. (a) CLDA. (b) SCLUDA. (c) TAADA. (d) DACD. (e) OSBP. (f) DAMC. (g) MTS. (h) STA. (i) UADAL. (j) OMEGA. (k) ANNA. (l) WGDT.

and Unk of the proposed method reached 88.2% and 90.2%, respectively, and the HOS reached 88.9%. HOS improves by 26.2% compared to other best methods. The classification maps of the proposed method and compared methods are shown in Fig. 12.

E. Ablation

In order to verify the effectiveness of the modules of the proposed method, ablation experiments were performed on four tasks. The results of the ablation experiment are shown in Table IX. We evaluate four methods:

1) using CA, domain adversarial learning (DA), and dynamic thresholding using MSE loss (DTM);
2) using CA, DA, and dynamic thresholding learning (DT);
3) using CA, weighted-domain adversarial learning (WDA), and DTM;
4) the proposed method WGDT, using CA, WDA, and DT.

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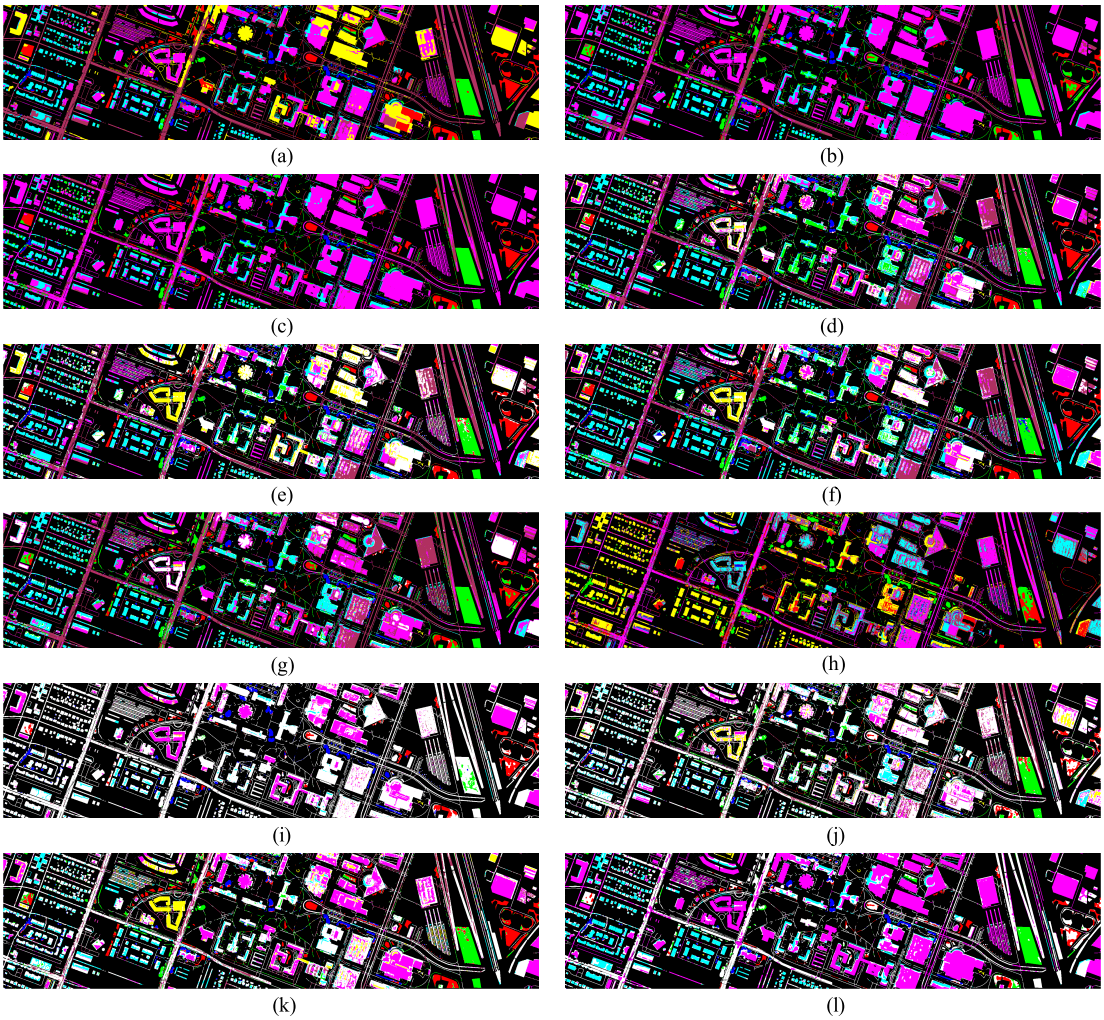


Fig. 10. Classification map of the HU13–HU18 task. (a) CLDA. (b) SCLUDA. (c) TAADA. (d) DACD. (e) OSBP. (f) DAMC. (g) MTS. (h) STA. (i) UADAL. (j) OMEGA. (k) ANNA. (l) WGDT.

TABLE IX
RESULTS ON THE PU–PC, HU13–HU18, DIONI–LOUKIA, AND ZY–GF TASKS IN THE ABLATION STUDY

Method	PU-PC	HU13-HU18	Dioni-Loukia	ZY-GF
CA + DA + DTM	56.0±13.0	53.6±3.9	45.1±0.8	71.5±18.9
CA + DA + DT	63.9±10.1	60.3±3.6	48.4±1.4	85.1±7.9
CA + WDA + DTM	72.6±7.0	54.1±2.4	45.7±0.9	72.7±12.8
WGDT	**76.5±4.6**	**61.1±2.8**	**48.9±1.3**	**88.9±3.8**

Compared with CA + DA + DTM, HOS improves by 20.5%, 7.5%, 3.8%, and 17.4% on PU–PC, HU13–HU18, Dioni–Loukia, and ZY–GF tasks, respectively. Through ablation study, the effectiveness of each part of the proposed method is demonstrated.

F. Parameter Discussion

In order to explore the effect of different hyperparameters on the performance of the proposed model, the grid search method is used to conduct experiments. Specifically, different values of the hyperparameters α, β, λ, φ, and m are set, respectively, and the results of the test after the model is trained under different hyperparameters are recorded. By comparison, the best hyperparameters are obtained.

1) Adjust the values of $\alpha \in \{0.001, 0.01, 0.05, 0.10, 0.5, 1.0, 1.5, 2.0\}$ and fix other hyperparameters. As shown in Fig. 13(a), as α increases, HOS shows an overall trend of first increasing and then decreasing. On the PU–PC, Dioni–Loukia, and ZY–GF tasks, HOS reaches the maximum value when $\alpha = 0.01$. On the HU13–HU18 task, when $\alpha = 0.1$, HOS reaches the maximum value.
2) Adjust the values of $\beta \in \{0.05, 0.10, 0.5, 1.0, 1.5, 2.0\}$ and fix other hyperparameters. As shown in Fig. 13(b), as β increases, on the PU–PC task, HOS shows a trend of first significantly increasing and then stabilizing. When $\beta = 1.5$, HOS reaches the maximum value. On the HU13–HU18 and ZY–GF tasks, HOS showed a trend of first significantly increasing and then decreasing. When $\beta = 1.0$, HOS reached the maximum value. while on the Dioni–Loukia task, the change in HOS was not obvious. and HOS reaches the maximum value when $\beta = 1.0$.
3) Adjust the values of $\lambda \in \{0.05, 0.10, 0.5, 1.0, 1.5, 2.0\}$ and fix other hyperparameters. As shown in Fig. 13(c), as λ increases, the change of HOS is not obvious on

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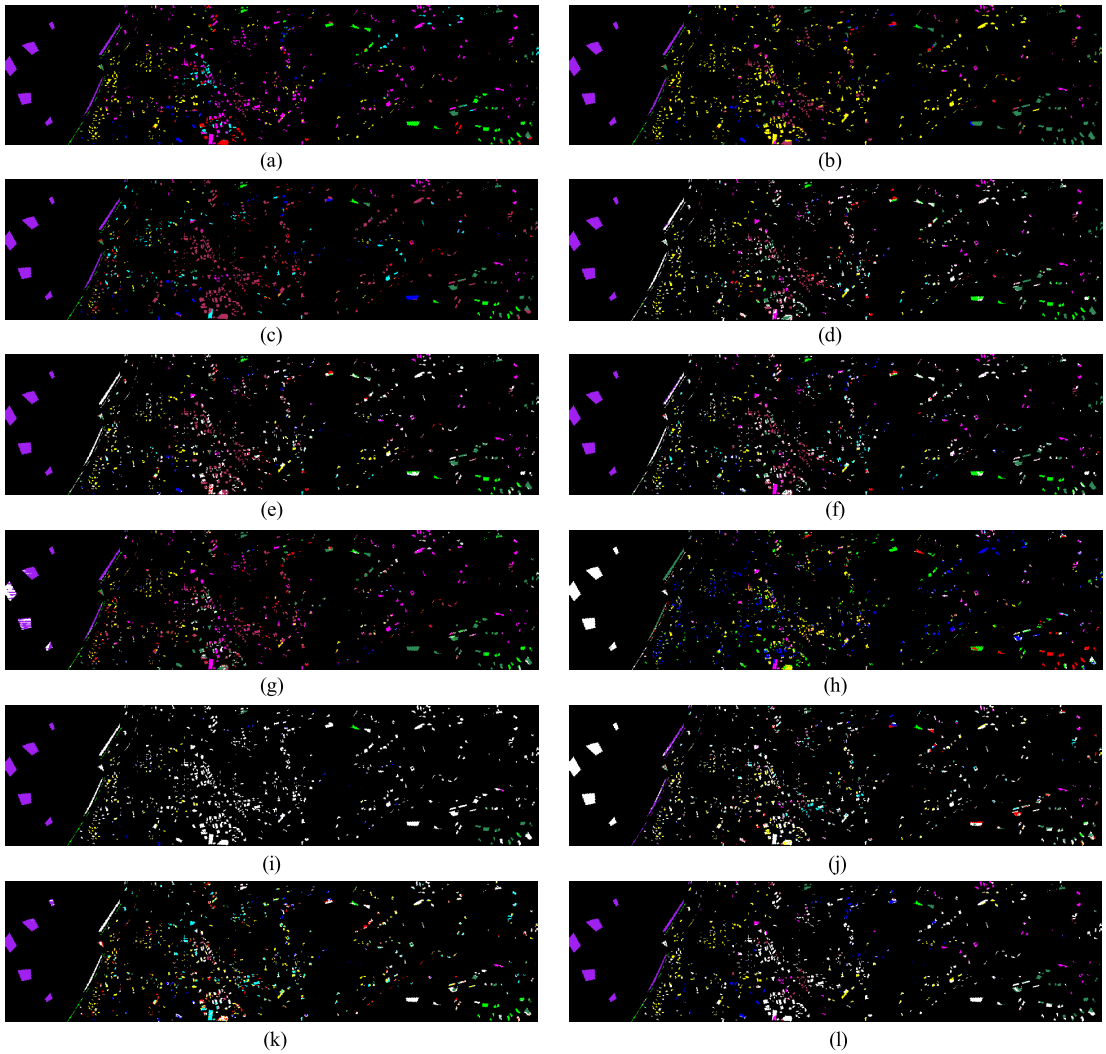


Fig. 11. Classification map of the Dioni–Loukia task. (a) CLDA. (b) SCLUDA. (c) TAADA. (d) DACD. (e) OSBP. (f) DAMC. (g) MTS. (h) STA. (i) UADAL. (j) OMEGA. (k) ANNA. (l) WGDT.

the PU–PC task. And when $\lambda = 0.1$, HOS reaches the maximum value. On the HU13–HU18 task, HOS shows a trend of first increasing and then decreasing and stabilizing, and HOS reaches the maximum value when $\lambda = 0.1$. On the Dioni–Loukia task, HOS shows a trend of first increasing and then leveling off, and when $\lambda = 1.0$, HOS reaches the maximum value. On the ZY–GF task, HOS shows a trend of first increasing and then decreasing and stabilizing, and when $\lambda = 0.5$, HOS reaches the maximum value.

4) Adjust the values of $\varphi \in \{6, 8, 10, 12, 14\}$ and fix other hyperparameters. As shown in Fig. 13(d), as φ increases, on the PU–PC task, HOS reaches the maximum value when $\varphi = 12$. On the HU13–HU18 and ZY–GF tasks, HOS showed a trend of first significantly increasing and then decreasing, and when $\varphi = 10$, HOS reached the maximum value. On the Dioni–Loukia task, HOS reached the maximum value when $\varphi = 10$.
5) Adjust the values of $m \in \{0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.5\}$ and fix other hyperparameters. As shown in Fig. 13(e), as m increases, HOS is relatively stable on the PU–PC, HU13–HU18, and Dioni–Loukia, and reaches the maximum value when $m = 0.3$, 0.01, and 0.2, respectively. On the ZY–GF task, HOS shows a trend of first stabilizing and then decreasing, and when $m = 0.005$, HOS reaches the maximum value.

Considering the above situations, the model can not only accurately classify known class samples in the target domain, but also accurately identify unknown class samples. Therefore, on the PU–PC task, the hyperparameters α, β, λ, φ, and m are set to 0.01, 1.5, 0.1, 12, and 0.3, respectively; on the HU13–HU18 task, the hyperparameters are set to 0.1, 1.0, 0.1, 10, and 0.01, respectively; on the Dioni–Loukia task, the hyperparameters are set to 0.01, 1.0, 1.0, 10, and 0.2, respectively; and on the ZY–GF task, the hyperparameters are set to 0.01, 1.0, 0.5, 10, and 0.005, respectively.

G. Computational Complexity

To evaluate the computational complexity of the proposed method, the training time, testing time, the number of floating point operations (FLOPs), and the number of parameters of

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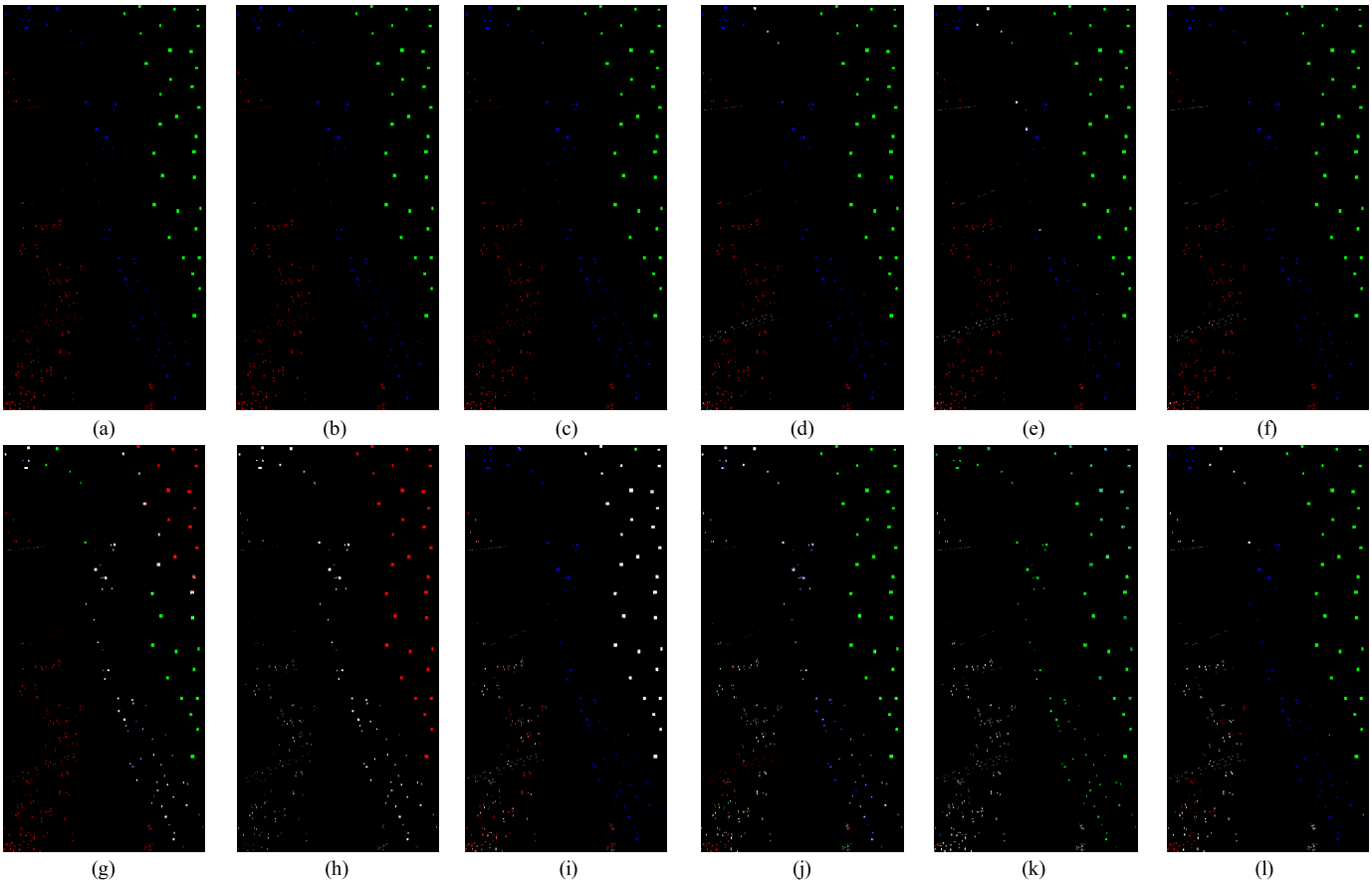


Fig. 12. Classification map of the ZY–GF task. (a) CLDA. (b) SCLUDA. (c) TAADA. (d) DACD. (e) OSBP. (f) DAMC. (g) MTS. (h) STA. (i) UADAL. (j) OMEGA. (k) ANNA. (l) WGDT.

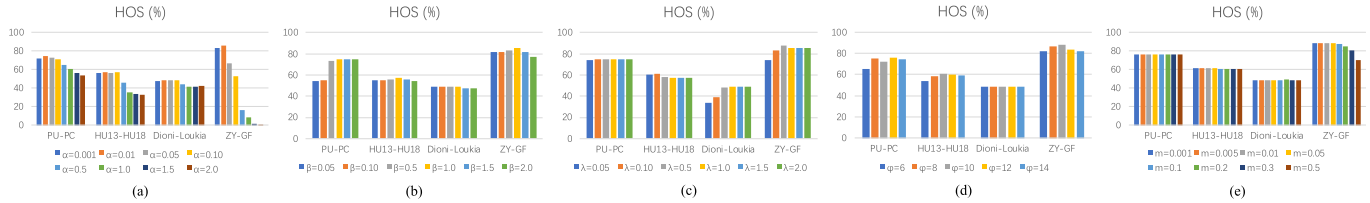


Fig. 13. Parameter discussion. The grid search method is used to find the optimal values of hyperparameters α, β, λ, φ, and m. (a) Adjust the value of α. (b) Adjust the value of β. (c) Adjust the value of λ. (d) Adjust the value of φ. (e) Adjust the value of m.

TABLE X

COMPUTATION TIME (S), FLOPS, AND PARAMETERS ON EACH TASK WITH DIFFERENT METHODS

		Closed-set			Open-set								
		CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	MTS	STA	UADAL	OMEGA	ANNA	WGDT
PU-PC	training time	397.41	361.63	167.29	277.82	384.42	350.31	2060.77	296.29	2187.91	500.25	770.09	312.11
	testing time	8.90	3.85	14.41	7.66	12.03	6.16	24.98	50.15	28.77	6.02	54.90	13.22
	FLOPS	453M	6263M	9843M	203M	199M	730M	52112M	26047M	5089M	6301M	56862M	8394M
	parameters	194K	1876K	770K	330K	316K	1439K	57368K	27929K	48853K	23525K	32976K	1976K
HU13-HU18	training time	1715.36	3132.34	96.12	242.41	421.22	307.04	2115.01	308.98	10117.76	635.01	679.24	263.86
	testing time	46.57	15.50	91.26	33.68	70.63	34.13	153.70	234.79	186.81	32.42	299.33	77.13
	FLOPS	419M	2917M	4915M	105M	102M	583M	52076M	26028M	5076M	6283M	56860M	3932M
	parameters	184K	1750K	686K	322K	308K	1431K	57365K	27928K	48850K	23523K	32975K	1850K
Dioni-Loukia	training time	313.42	790.69	713.28	296.61	603.00	367.35	2186.23	317.41	1009.43	611.95	997.27	605.11
	testing time	1.27	0.78	2.09	0.96	0.95	0.96	4.05	6.63	3.78	1.29	7.18	1.87
	FLOPS	500M	10850M	16596M	336M	333M	931M	52298M	26072M	5109M	6327M	56872M	14510M
	parameters	209K	2050K	886K	340K	327K	1452K	58424K	27932K	48866K	23531K	32980K	2150K
ZY-GF	training time	50.82	148.26	51.23	200.03	361.75	224.25	1102.96	283.63	594.86	578.16	311.92	367.30
	testing time	0.82	0.66	0.71	0.52	0.50	0.51	1.41	2.99	3.33	0.73	3.59	0.67
	FLOPS	481M	9114M	14040M	284M	280M	850M	51871M	26061M	5100M	6314M	56851M	3048M
	parameters	203K	1981K	841K	336K	322K	1441K	55267K	27930K	49364K	23518K	32973K	2078K

several methods were compared. The results are shown in Table X. MTS adopts a training method in which two stages alternate with each other until convergence, resulting in longer training time. UADAL uses posterior reasoning to distinguish

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known classes from unknown classes, and needs to fit a beta distribution on the target domain, which requires longer training and testing time. This is even more problematic in HU13–HU18, which has more samples in the target domain. Significantly, compared with several other methods, although the proposed method has no significant advantages in training and testing times, it can offer better performance in classification tasks.

V. Conclusion

In this article, an OSDA based on the WGDT framework is proposed, which can effectively alleviate domain differences and identify unknown class samples in the target domain. The anchor-based weight calculation method is used to provide reliable weights for domain adversarial learning and alleviate the negative transfer phenomenon during domain alignment. In addition, a learnable dynamic thresholding is proposed to automatically determine the decision boundary of known classes in the target domain and be used to accurately identify unknown classes. Our method is validated on four tasks consisting of eight datasets, and the experimental results show that it can outperform other state-of-the-art methods.

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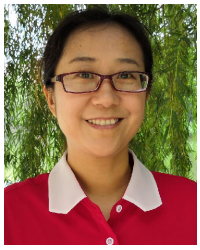
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